18

Using Multiples

While discounted cash flow (DCF) is the most accurate and flexible method for valuing companies, using a relative valuation approach, such as juxtaposing the earnings multiples of comparable companies, can provide insights and help you summarize and test your valuation. In practice, however, multiples are often used in a superficial way that leads to erroneous conclusions. This chapter explains how to use multiples correctly. Most of the focus will be on earnings multiples, the most commonly used variety. At the end, we'll also touch on some other multiples.

The basic idea behind using multiples for valuation is that similar assets should sell for similar prices, whether they are houses or shares of stock. In the case of a share of stock, the typical benchmark is some measure of earnings, most popularly the price-to-earnings (P/E) multiple, which is simply the equity value of the company divided by its net income. Multiples can be used to value nontraded companies or divisions of traded companies and to see how a listed company is valued relative to peers. Companies in the same industry and with similar performance should trade at the same multiple.

Valuing a company by using multiples may seem straightforward, but arriving at useful insights requires careful analysis. Exhibit 18.1 illustrates what happens if you don't go deep enough in your multiples analysis. The managers of Company A, a producer of packaged foods, looked only at P/Es and were concerned that their company was trading at a P/E of 7.3 times while most of their peers were trading at a P/E of about 14, a discount of 50 percent. The management team believed the market didn't understand Company A's strategy or performance. In fact, management didn't understand the math of multiples. If the managers had looked at the more instructive multiple shown in the exhibit—net enterprise value to earnings before interest, taxes, and amortization (EV/EBITA)—they would have seen that the company was

EXHIBIT 18.1 **Multiples for Packaged Foods Companies**

$ billion

Company	Market value of equity	Enterprise value (equity + debt)	Net income (1 year forward)	EBITA (1 year forward)	Multiples: Price/ earnings	Multiples: Enterprise value/EBITA
A	2,783	9,940	381	929	7.3	10.7
B	13,186	16,279	856	1,428	15.4	11.4
C	8,973	11,217	665	1,089	13.5	10.3
D	14,851	22,501	1,053	2,009	14.1	11.2
Mean					12.6	10.9
Median					13.8	11.0
Mean (excluding A)					14.3	11.0
Median (excluding A)					14.1	11.2

trading right in line with its peers. The reason for the difference was that their company had much more debt relative to equity than the other companies. We estimated that if the company had had the same relative debt as its peers, its P/E also would have been 14. Except for very-high-growth companies, a company with higher debt relative to peers will have a lower P/E because more debt translates to higher risk for shareholders and a higher cost of equity. Therefore, each dollar of earnings (and cash flow to shareholders) will be worth less to an investor.[1]

To use earnings multiples properly, you should dig into the accounting statements to make sure you are comparing companies on an apples-to-apples basis. You also must choose the right companies to compare. Keep in mind these five principles for correctly using earnings multiples:

1. *Value multibusiness companies as a sum of their parts.* Even companies that appear to be in a single industry will often compete in subindustries or product areas with widely varying return on invested capital (ROIC) and growth, leading to substantial variations in multiples.
2. *Use forward estimates of earnings.* Multiples using forward earnings estimates typically have much lower variation across peers, leading to a narrower range of uncertainty of value. They also embed future expectations better than multiples based on historical data.
3. *Use the right multiple, usually net enterprise value to EBITA or net enterprise value to NOPAT.* Although the P/E is widely used, it is distorted by capital structure and nonoperating gains and losses. (In this book, when we

[1] The P/E multiple is a function of return on capital, cost of capital, and growth. For very-high-growth companies, whose enterprise multiples are greater than the multiple for debt, the multiple will actually increase with leverage. See also Appendix D.

refer to the enterprise value multiple, including abbreviations such as EV/EBITA, we use "enterprise value" as shorthand for net enterprise value, equal to the value of operations.)

4. *Adjust the multiple for nonoperating items.* Nonoperating items embedded in reported EBITA, as well as balance sheet items like excess cash and pension items, can lead to large distortions of multiples.
5. *Use the right peer group, not a broad industry average.* A good peer group consists of companies that not only operate in the same industry but also have similar prospects for ROIC and growth.

VALUE MULTIBUSINESS COMPANIES AS A SUM OF THEIR PARTS

Most large companies, even if they operate in a single industry, have business units that are in subindustries with different competitive dynamics and therefore differ widely in ROIC and growth. For example, many analysts would classify Johnson & Johnson as a health-care company, but its three major units (pharmaceuticals, medical devices, and consumer health products) have widely varying economic characteristics in terms of growth and return on capital. Each of the units will therefore have different valuation multiples. For such multibusiness companies, a valuation using multiples requires a sum-of-parts approach, which values each business unit with a multiple appropriate to its peers and performance.

Even companies in more narrowly defined sectors often have units with different economics. For example, oil and gas services companies provide oil and gas companies with equipment and services that might include bottom hole assemblies, drill pipes, pressure-control services, intervention services, pressure pumping, fluid handling, subsea construction, and even temporary housing for workers. Some of these product areas, including bottom hole assemblies and drill pipes, tend to earn much higher returns on capital than pressure-control services and intervention services. Ideally, you would value units by using as fine-grained an approach as possible, comparing them with companies that have similar units and economics.

For an example of a good sum-of-parts valuation, see Exhibit 18.2. For each unit of this disguised company, we apply a different multiple to its earnings based on different peers. Then we sum the values of the units to estimate net enterprise value. To estimate equity value, we add nonoperating assets and subtract debt and debt equivalents.

Note that the business unit multiples range from the midteens to below ten. Without the sum-of-parts approach, it would be impossible to value this company accurately. We also used ranges for the value of each unit, reflecting the imprecision of valuing any business based on the valuation of peers at a single point in time.

EXHIBIT 18.2 **Sample Sum-of-Parts Valuation**

	NOPAT, 2014, $ million	EV/NOPAT, times		Value, $ million	
		High	Low	High	Low
Business Unit 1	410	16.0	14.5	6,568	5,952
Business Unit 2	299	13.9	12.5	4,165	3,749
Business Unit 3	504	13.1	12.5	6,597	6,306
Business Unit 4	587	9.7	9.4	5,681	5,533
Business Unit 5	596	9.0	8.0	5,365	4,769
Business Unit 6	116	8.0	7.0	931	814
Corporate	(542)	8.0	9.1	(4,339)	(4,917)
Net Enterprise Value	1,971	12.7	11.3	24,968	22,207

	After-tax net income, 2013, $ million	Book value, $ million	Earnings multiple, 2013 times	Market value/ book value, times	Value, $ million	
					High	Low
Joint ventures	157	675	12.0	2.5	1,879	1,688
Other investments		1,525			1,525	1,525
Cash and marketable securities		2,879			2,879	2,879
Gross enterprise value					31,251	28,298
Debt		(10,776)			(10,776)	(10,776)
Unfunded retirement liabilities		(2,907)			(2,907)	(2,907)
Noncontrolling interest	(45)	(296)	12.0	2.5	(540)	(739)
Other		(1,940)			(1,940)	(1,940)
Equity value					15,088	11,937
Shares outstanding, millions					500	500
Equity value per share					$30.18	$23.87

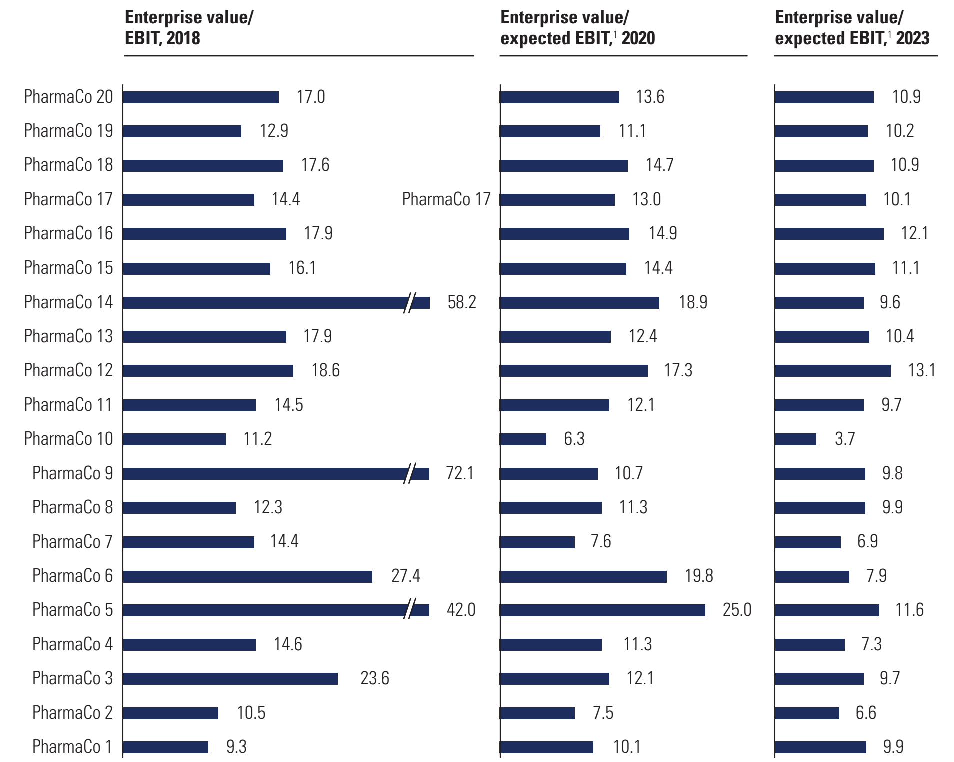
USE FORWARD EARNINGS ESTIMATES

When you are building multiples, the denominator should be a forecast of profits, preferably normalized for unusual items, rather than historical profits. Unlike backward-looking multiples, forward-looking multiples are consistent with the principles of valuation—in particular, that a company's value equals the present value of future cash flows, not sunk costs. When companies have recently acquired or divested significant parts of their operations, historical profits are even less meaningful. Normalized earnings estimates better reflect long-term cash flows by avoiding one-time items. For example, Warren Buffett and other disciples of value-investing guru Benjamin Graham don't use reported earnings. Rather, they rely on a sustainable level of earnings that they refer to as "earnings power."[2]

Forward-looking multiples generally also have lower variation across peer companies. A particularly striking example is the stock market valuation of the 20 largest pharmaceutical companies worldwide in 2019. The

[2] B. C. N. Greenwald, J. Kahn, P. D. Sonkin, and M. van Biema, *Value Investing: From Graham to Buffett and Beyond* (Hoboken, NJ: John Wiley & Sons, 2001).

EXHIBIT 18.3 **Pharmaceuticals: Backward- and Forward-Looking Multiples, September 2019**



[1] Consensus analyst forecast.

backward-looking ratio of enterprise value of last year's EBIT ranged from about 10 to more than 70 times (see Exhibit 18.3). The ratio of enterprise value to the next year's expected EBIT, based on equity analyst estimates, also showed significant variation, ranging from about 6 to 25 times. But when we extended the forecast window to four years, the variation across companies was significantly lower, with multiples for all but one company between about 7 and 12 times.

The convergence of multiples four years out in the pharmaceuticals industry is extreme. This is most likely due to the market's ability to project near-term earnings well, because drug introductions and patent expirations are well known. By contrast, it is difficult to differentiate long-term success across companies because it depends on an individual company's ability to discover or develop new drugs. No one has figured out a good way to do that.

Empirical evidence shows that forward-looking multiples are indeed more accurate predictors of value than historical multiples are. One empirical study examined the characteristics and performance of historical multiples versus forward industry multiples for a large sample of companies trading on U.S. exchanges.[3] When multiples for individual companies were compared with their industry multiples, their historical earnings-to-price (E/P) ratios had 1.6 times the standard deviation of one-year-forward E/P ratios (6.0 percent versus 3.7 percent). Other research, which used multiples to predict the prices of 142 initial public offerings, also found that multiples based on forecast earnings outperformed those based on historical earnings.[4] As the analysis moved from multiples based on historical earnings to multiples based on one- and two-year forecasts, the average pricing error fell from 55.0 percent to 43.7 percent to 28.5 percent, respectively, and the percentage of firms valued within 15 percent of their actual trading multiple increased from 15.4 percent to 18.9 percent to 36.4 percent.

To build a forward-looking multiple, choose a forecast year for EBITA that best represents the long-term prospects of the business. In periods of stable growth and profitability, next year's estimate will suffice. For companies generating extraordinary earnings (either too high or too low) or for companies whose performance is expected to change, use projections further out.

USE NET ENTERPRISE VALUE DIVIDED BY ADJUSTED EBITA OR NOPAT

Most financial websites and newspapers quote a price-to-earnings ratio by dividing a company's share price by the prior 12 months' GAAP-reported earnings per share. Yet these days, sophisticated investors and bankers use what we call forward-looking multiples of net enterprise value to EBITA (or NOPAT). They find that these multiples provide a more apples-to-apples comparison of company values.

The reasons for using forward earnings are the same as the ones discussed in the previous section. Using net enterprise value to EBITA (or NOPAT) rather than a P/E eliminates the distorting effect of different capital structures, nonoperating assets, and nonoperating income statement items, such as the nonoperating portion of pension expense. Any item that isn't a helpful indicator of a company's future cash-generating ability should be excluded from your calculation of the multiple. For example, one-time gains or losses and nonoperating expenses, such as the amortization of intangibles, have no direct relevance to future cash flows; including them in the multiple would distort comparisons with other companies.

[3] J. Liu, D. Nissim, and J. Thomas, "Equity Valuation Using Multiples," *Journal of Accounting Research* 40 (2002): 135–172.
[4] M. Kim and J. R. Ritter, "Valuing IPOs," *Journal of Financial Economics* 53, no. 3 (1999): 409–437.

Sometimes analysts use an alternative multiple: enterprise value to earnings before interest, taxes, depreciation, and amortization (EBITDA). Later in this section, we'll explain the logic of using EBITA or NOPAT instead of EBITDA.

Why Not Price to Earnings?

This book has focused throughout on the drivers of operating performance—ROIC, growth, and free cash flow—because the traditional metrics, such as return on assets (ROA) and return on equity (ROE), mix the effects of operations and capital structure. The same logic holds for multiples. Since the price-to-earnings ratio mixes capital structure and nonoperating items with expectations of operating performance, a comparison of P/Es is a less reliable guide to companies' relative value than a comparison of enterprise value (EV) to EBITA or NOPAT.

To show how capital structure distorts the P/E, Exhibit 18.4 presents financial data for four companies, named A through D. Companies A and B trade at 10 times enterprise value to EBITA, and Companies C and D trade at 25 times enterprise value to EBITA. In each pair, the companies have different P/Es. Companies A and B differ only in how their business is financed, not in their operating performance. The same is true for Companies C and D.

Since Companies A and B trade at typical enterprise value multiples, the P/E drops for the company with higher leverage. This is because the EV-to-EBITA ratio ($1,000 million/$100 million = 10 times) is lower than the ratio of debt value to interest expense ($400 million/$20 million = 20 times).

EXHIBIT 18.4 **P/E Multiple Distorted by Capital Structure**

$ million

	Company A	Company B	Company C	Company D
Income statement				
EBITA	100	100	100	100
Interest expense	–	(20)	–	(25)
Earnings before taxes	100	80	100	75
Taxes	(40)	(32)	(40)	(30)
Net income	60	48	60	45
Market values				
Debt	–	400	–	500
Equity	1,000	600	2,500	2,000
Enterprise value (EV)	1,000	1,000	2,500	2,500
Multiples, times				
EV/EBITA	10.0	10.0	25.0	25.0
Price/earnings	16.7	12.5	41.7	44.4

Since the blend of debt at 20 times and pretax equity must equal the enterprise value at 10 times, the pretax equity multiple must drop below 10 times to offset the greater weight placed on high-multiple debt.[5] The opposite is true when enterprise value to EBITA exceeds the ratio of debt to interest expense (less common, given today's low interest rates). Company D has a higher P/E than Company C because Company D uses more leverage than Company C. In this case, a high pretax P/E (greater than 25 times) must be blended with the debt multiple (20 times) to generate an EV-to-EBITA multiple of 25 times.

Why Not EV to EBIT?

It's clear that shifting to enterprise-value multiples provides better insights and comparisons across peer companies. The next question is what measure of operating profits to use in the denominator—EBIT, EBITDA, EBITA (adjusted), or NOPAT? We recommend EBITA or NOPAT.

The difference between EBIT and EBITA is amortization of intangible assets. Most often, the bulk of amortization is related to acquired intangible assets, such as customer lists or brand names. Chapter 11 explained why we exclude amortization of acquired intangibles from the calculation of ROIC and free cash flow. It is noncash, and, unlike depreciation of physical assets, the replacement of these intangible assets is already incorporated in EBITA through line items such as marketing and selling expenses. So using EBITA is preferred, both from a logical perspective and because it leads to more comparable multiples across peers.

To illustrate the distortion caused by amortization of acquired intangible assets, we compare two companies with the same size and underlying operating profitability. The difference is that Company A achieved its current size by acquiring Company B, whereas Company C grew organically. Exhibit 18.5 compares these companies before and after A's acquisition of B.

Concerned that its smaller size might lead to a competitive disadvantage, Company A purchased Company B. Assuming no synergies, the combined financial statements of Companies A and B are identical to Company C's with two exceptions: acquired intangibles and amortization. Acquired intangibles are recognized when a company is purchased for more than its book value. In this case, Company A purchased Company B for $1,000 million, which is $750 million greater than its book value. If these acquired intangibles are separable and identifiable, such as patents, Company A + B must amortize them over the estimated life of the asset. Assuming an asset life of ten years, Company A + B will record $75 million in amortization each year.

[5] Appendix D derives the explicit relationship between a company's actual P/E and its unlevered P/E, that is, the P/E as if the company were entirely financed with equity. For companies with large unlevered P/Es (i.e., companies with significant opportunities for future value creation), P/E systematically increases with leverage. Conversely, companies with small unlevered P/Es would exhibit a drop in P/E as leverage rises.

EXHIBIT 18.5 **Enterprise-Value-to-EBIT Multiple Distorted by Acquisition Accounting**

$ million

	Before acquisition			After A acquires B	
	Company A	**Company B**	**Company C**	**Company A + B**	**Company C**
EBIT					
Revenues	375	125	500	500	500
Cost of sales	(150)	(50)	(200)	(200)	(200)
Depreciation	(75)	(25)	(100)	(100)	(100)
EBITA	150	50	200	200	200
Amortization	–	–	–	(75)	–
EBIT	150	50	200	125	200
Invested capital					
Organic capital	750	250	1,000	1,000	1,000
Acquired intangibles	–	–	–	750	–
Invested capital	750	250	1,000	1,750	1,000
Enterprise value	1,500	500	2,000	2,000	2,000
Multiples, times					
EV/EBITA	10.0	10.0	10.0	10.0	10.0
EV/EBIT	10.0	10.0	10.0	16.0	10.0

The bottom of Exhibit 18.5 reports enterprise value multiples using EBITA and EBIT, both before and after the acquisition. Since all three companies generated the same level of operating performance, they traded at identical multiples before the acquisition, 10 times EBIT (and EBITA). After the acquisition, the combined Company A + B should continue to trade at a multiple of 10 times EBITA, because its performance is identical to that of Company C. However, amortization expense causes EBIT to drop for the combined company, so its EV-to-EBIT multiple increases to 16 times. This rise in the multiple does not reflect a premium, however (remember, no synergies were created). It is merely an accounting artifact. Companies that acquire other companies must recognize amortization, whereas companies that grow organically have none to recognize. To avoid forming a distorted picture of their relative operating performance, use EV-to-EBITA multiples.

In limited cases, companies will capitalize organic investments in intangible assets. For example, telecommunication service providers capitalize the purchase costs for spectrum licenses and then amortize them over their useful life. In a similar way, development costs for software that is to be sold or licensed to third parties can be capitalized and amortized under IFRS and U.S. GAAP if certain conditions are met. In such cases, the amortization charges are operating costs and should be separated from acquisition amortization. Just like depreciation charges, operating amortization should be included in adjusted EBITA.

EXHIBIT 18.6 **Enterprise-Value-to-EBITDA Multiple Distorted by Capital Investment**

$ million

	Company A	Company B
Income statement		
Revenues	1,000	1,000
Raw materials	(100)	(250)
Operating costs	(400)	(400)
EBITDA	500	350
Depreciation	(200)	(50)
EBITA	300	300
Operating taxes	(90)	(90)
NOPAT	210	210
Multiples, times		
EV/EBITA	10.0	10.0
EV/EBITDA	6.0	8.6

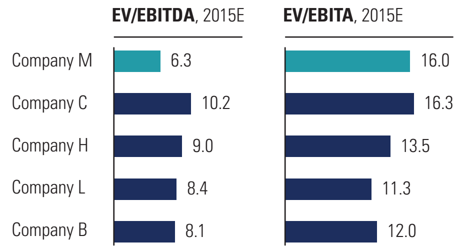
	Company A	Company B
Free cash flow		
NOPAT	210	210
Depreciation	200	50
Gross cash flow	410	260
Investment in working capital	(60)	(60)
Capital expenditures	(200)	(50)
Free cash flow	150	150
Enterprise value	3,000	3,000

Choosing between EBITA and EBITDA

A common alternative to the EBITA multiple is the EBITDA multiple. Many practitioners use EBITDA multiples because depreciation is, strictly speaking, a noncash expense, reflecting sunk costs, not future investment. This logic, however, does not apply uniformly. For many industries, depreciation of existing assets is the accounting equivalent of setting aside the future capital expenditure that will be required to replace the assets. Subtracting depreciation from the earnings of such companies therefore better represents future cash flow and consequently the company's valuation.

To see this, consider two companies that differ in only one aspect: in-house versus outsourced production. Company A manufactures its products using its own equipment, whereas Company B outsources manufacturing to a supplier. Exhibit 18.6 provides financial data for each company. Since Company A owns its equipment, it recognizes significant annual depreciation—in this case, $200 million. Company B has less equipment, so its depreciation is only $50 million. However, Company B's supplier will include its own depreciation costs in its price, and Company B will consequently pay more for its raw materials. Because of this difference, Company B generates EBITDA of only $350 million, versus $500 million for Company A. This difference in EBITDA will lead to differing multiples: 6.0 times for Company A versus 8.6 times for Company B. Does this mean Company B trades at a valuation premium? No, when Company A's depreciation is deducted from its earnings, both companies trade at 10.0 times EBITA.

EXHIBIT 18.7 **Company M Peer Multiples Comparison**



When computing the EV-to-EBITDA multiple in the previous example, we failed to recognize that Company A (the company that owns its equipment) will have to expend cash to replace aging equipment: $200 million for Company A versus $50 million for Company B (see the right side of Exhibit 18.6). Since capital expenditures are recorded in free cash flow and not NOPAT, the EBITDA multiple is distorted.

We came across an interesting example in a processing industry, as shown in Exhibit 18.7. On an EV-to-EBITDA basis, Company M trades at a multiple of 6.3 times, far below its peers' multiples of 8.1 to 10.2 times. However, on an EV-to-EBITA basis, it actually trades at the high end of its peers. In this industry, companies have to replace depreciated assets constantly, so the EBITA multiple provides a better comparison of valuation levels. In Company H's case, its low cash margins also contribute to the larger gap between EBITA and EBITDA.

In some situations, EBITDA scales a company's valuation better than EBITA. These occur when current depreciation is not an accurate predictor of future capital expenditures. For instance, consider two companies, each of which owns a machine that produces identical products. Both machines have the same cash-based operating costs, and each company's products sell for the same price. If one company paid more for its equipment (for whatever reason—perhaps poor negotiation), it will have higher depreciation and, thus, lower EBITA. Valuation, however, is based on future discounted cash flow, not past profits. And since both companies have identical cash flow, they should have identical values.[6] We would therefore expect the two companies to have identical multiples. Yet, because EBITA differs across the two companies, their multiples will differ as well.

[6] Since depreciation is tax deductible, a company with higher depreciation will have a smaller tax burden. Lower taxes lead to higher cash flows and a higher valuation. Therefore, even companies with identical EBITDAs will have different EBITDA multiples. The distortion, however, is less pronounced.

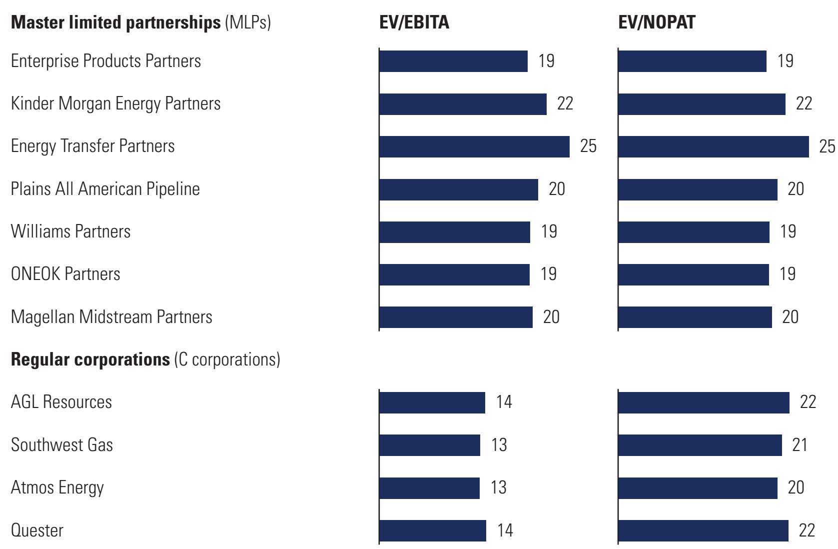
NOPAT vs. EBITA

Analysts and investors often use enterprise value to EBITA instead of NOPAT because there is no need to figure out the operating taxes on EBITA. (Reported taxes are not usually a good predictor of operating taxes, because they include nonoperating items. Therefore, most analysts ignore taxes altogether.) We often use EBITA because it's common practice and works well when all the companies in the peer group have the same operating tax rate, as when they all operate within a single tax jurisdiction. However, when tax rates are different, NOPAT is a better measure to use.

U.S. oil and gas pipeline companies provide a classic example. Until the mid-2010s, many pipeline companies were organized as master limited partnerships (MLPs), which eliminated an entire layer of taxation compared with pipeline companies organized as regular corporations, called C corporations in the U.S. tax code. Unlike C corporations, MLPs pay no corporate income taxes; rather, investors pay taxes on their share of the profits. Exhibit 18.8 shows that the stock market clearly reflected these tax differences when valuing companies in these industries. NOPAT multiples across all the companies are in a narrow range of 19 to 25 times. The EBITA multiples, however, show a clear delineation between the regular corporations and the MLPs. While the EBITA multiples for the MLPs remain the same, at 19 to 25 times, the multiples for regular corporations drop to 13 to 14 times. Clearly, the NOPAT multiples are superior in this case.

EXHIBIT 18.8 **Enterprise Value to EBITA vs. Enterprise Value to NOPAT**

U.S. pipeline companies, June 2013

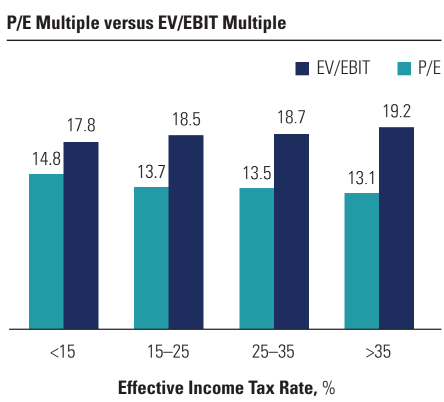


The stock market recognizes differences in tax rates not only for pipeline companies but across all sectors. The difference between a company's post- and pretax earnings valuation multiple simply follows from the company's tax rate. If the stock market correctly reflects taxation in company valuations, we would expect that for companies with higher tax rates, the difference between their pre- and posttax earnings multiples also would be bigger. This is indeed the pattern that we found when examining the market valuations of the largest U.S. companies between 2013 and 2017 (before the Tax Cuts and Jobs Act of 2017). Exhibit 18.9 shows the average difference in pretax earnings multiples (EV/EBIT) and posttax earnings multiples (P/E) over the five-year period for companies categorized according to their income tax rates. As predicted, the difference between the multiples steadily increases with the tax rate that a company pays. Differences in tax rates clearly matter for market valuation. Thus, when companies face different tax rates, they should not be valued at the same EBITA multiple (or any other pretax earnings multiple).

This is an important consideration for international comparisons, because corporate tax rates vary widely from country to country. For example, as of 2019, the Irish corporate tax rate is one of the lowest, at 12.5 percent, the U.S. tax rate is at 21.0 percent, and the French tax rate is one of the highest, at 34.4 percent. Because of such variations, companies in the same industry with a different geographic mix of operations can have different tax rates, which must be factored into their valuation using multiples. If the tax rates are different across peers, use net enterprise value to NOPAT rather than net enterprise value to EBITA.

EXHIBIT 18.9 **Difference between Pre- and Posttax Earnings Multiples for U.S. Stock Market**

2013–2017, average



Source: S&P CapitaliQ

ADJUST FOR NONOPERATING ITEMS

In a presentation to a group of professional investors, we provided the audience with financial data on two companies. We then asked the audience which company traded at a higher EV multiple. The results were surprising. Upon polling the group, we discovered that there was no common agreement on how to compute the EV multiple. A group of 100 professionals generated nearly a dozen different comparisons. Further investigation revealed that the primary cause of this divergence was inconsistency in defining enterprise value.

Only one approach to building an EV-to-EBITA multiple is theoretically consistent. Enterprise value should include *only* the portion of value attributable to assets and liabilities that generate (adjusted) EBITA. Strictly speaking, it should be referred to as "net" enterprise value, meaning net of nonoperating assets. Including value, such as the value of joint ventures, in the numerator without including its corresponding income or loss in the denominator will systematically distort the multiple upward. Conversely, failing to recognize a component of enterprise value will understate the numerator and bias the multiple downward. This occurs, for example, when the value of noncontrolling interest is not added to the value of common equity.

Oracle offers an example of a biased multiple. At the end of September 2019, Oracle had $38 billion of cash and marketable securities. With total debt, pensions, and debt equivalents of $76 billion, as well as equity of $177 billion, it had a gross enterprise value of $253 billion. Subtracting nonoperating cash gives a net enterprise value of $216 billion. With expected EBITA of $19 billion, its gross enterprise value to EBITA would be 13.3 times, while its net enterprise value to EBITA would be 11.3 times, or 15 percent lower.[7]

A way to think about the difference is to think of Oracle as a portfolio with two components: one is an operating business that sells software and services, and the other is a pile of cash. The operating business is valued at 11.3 times EBITA, while if the cash earned 1.0 percent pretax, it would be valued at 100 times (the inverse of the earnings yield). The company as a whole is valued at the weighted average of the two multiples, 13.3 times. Since the 13.3 times is a weighted average of two very different numbers, it doesn't provide any insight into how to think about Oracle's value.

To see how the math provides additional clarity, Exhibit 18.10 presents three companies—A, B, and C—with identical EV-to-EBITA multiples. Company A holds only core operating assets and is financed by traditional debt and equity. Its combined market value of debt and equity equals $900 million. Dividing $900 million by $100 million in EBITA leads to an EV multiple of 9 times.

[7] Even if we adjusted EBITA to include income on the cash (say, 1 percent after taxes), its gross enterprise value multiple would have been roughly the same.

EXHIBIT 18.10 **Enterprise Value Multiples and Complex Ownership**

$ million

	Company A	Company B	Company C
Partial income statement			
EBITA	100	100	100
Interest income	–	4	–
Interest expense	(18)	(18)	(18)
Earnings before taxes	82	86	82
Gross enterprise value			
Value of core operations	900	900	900
Excess cash	–	100	–
Nonconsolidated subsidiaries	–	200	–
Gross enterprise value	900	1,200	900
Debt	300	300	300
Noncontrolling interest	–	–	100
Market value of equity	600	900	500
Gross enterprise value	900	1,200	900
Multiples, times			
Net EV/EBITA	9.0	9.0	9.0
Debt plus equity minus cash/EBITA	9.0	11.0	8.0
Debt plus equity/EBITA	9.0	12.0	8.0

Company B operates a similar business to Company A but also owns $100 million in excess cash and a minority stake in a nonconsolidated subsidiary, valued at $200 million. Since excess cash and nonconsolidated subsidiaries do not contribute to EBITA, do not include them in the numerator of an EV-to-EBITA multiple. To compute a net enterprise value consistent with EBITA, sum the market value of debt and equity ($1,200 million), and subtract the market value of nonoperating assets ($300 million).[8] Divide the resulting net enterprise value ($900 million) by EBITA ($100 million). The result is an EV-to-EBITA multiple of 9, which matches that of Company A. Failing to subtract the market value of nonoperating assets will lead to a multiple that is too high. For instance, if you divide debt plus equity by EBITA for Company B, the resulting multiple is 12 times, three points higher than the correct value.

Similar adjustments are necessary for financial claims other than debt and equity. To calculate enterprise value consistently with EBITA, you must include the market value of all financial claims, not just debt and equity. For Company C, outside investors hold a noncontrolling interest in a consolidated subsidiary. Since the noncontrolling stake's value is supported by EBITA, you

[8] Alternatively, we could adjust the denominator rather than the numerator by adding interest income to EBITA. This definition of EV to EBITA is consistent but is biased upward. This is because the multiple for excess cash typically exceeds that of core operations. The greater the proportion of cash to overall value, the higher the resulting multiple.

must include it in the enterprise value calculation. Otherwise, the EV-to-EBITA multiple will be biased downward. For instance, when only debt plus equity is divided by EBITA for Company C, the resulting multiple is only 8 times.

As a general rule, any nonoperating asset that does not contribute to EBITA should be removed from enterprise value. This includes not only the market value of excess cash and nonconsolidated subsidiaries, as just mentioned, but also excess real estate, other investments, and the market value of prepaid pension assets. Financial claims include debt and equity, but also minority interest, the value of unfunded pension liabilities, and the value of employee grants outstanding. A detailed discussion of nonoperating assets and financial claims is presented in Chapter 16.

A trickier adjustment is needed for pensions and other retirement benefits, as explained in Chapter 23. Treat the unfunded liabilities as debt or the excess assets as a nonoperating asset. In addition, exclude the nonoperating parts of pension expense from EBITA.

USE THE RIGHT PEER GROUP

Selecting the right peer group is critical to coming up with a reasonable valuation using multiples. Common practice is to select a group of 8 to 15 peers and take the average of the multiples of the peers. Getting a reasonable valuation, though, requires judgment about which companies and their multiples are truly relevant for the valuation.

A common approach to identifying peers is to use the Standard Industrial Classification (SIC) codes or the newer Global Industry Classification Standard (GICS) system developed by Standard & Poor's and Morgan Stanley.[9] These may be a good starting point, but they are usually too broad for a good valuation analysis. For example, United Parcel Service (UPS) is included in the air freight and logistics GICS code, which includes dozens of companies, most of which do not compete with UPS in its core business of delivering small parcels. Another approach is to use peers provided by the company being valued. However, companies often provide aspirational peers rather than companies that truly compete head-to-head. It is better to have a smaller number of peers of companies that truly compete in the same markets with similar products and services.

Even if you find companies that compete head-to-head, differences in performance may justify differences in multiples. Remember the value driver formula expressed as a multiple:

[9] Beginning in 1997, SIC codes were replaced by a major revision called the North American Industry Classification System (NAICS). The NAICS six-digit code not only provides for newer industries but also reorganizes the categories on a production/process-oriented basis. The Securities and Exchange Commission (SEC), however, still lists companies by SIC code.

$$\frac{\text{Value}}{\text{EBITA}} = \frac{(1-T)\left(1-\frac{g}{\text{ROIC}}\right)}{\text{WACC}-g}$$

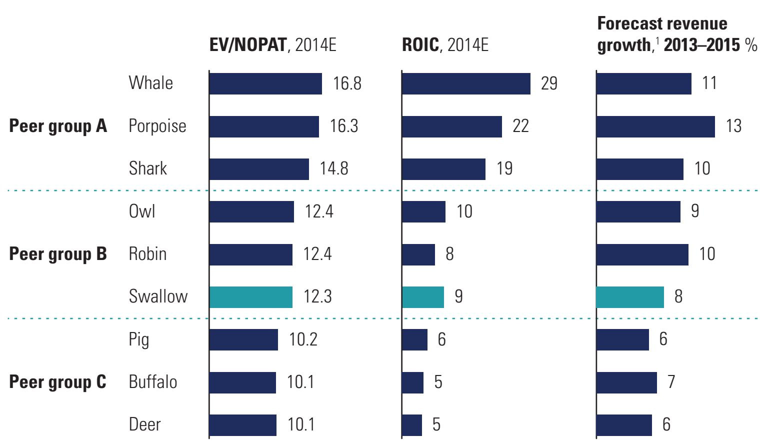
or

$$\frac{\text{Value}}{\text{NOPAT}} = \frac{\left(1-\frac{g}{\text{ROIC}}\right)}{\text{WACC}-g}$$

As both versions of the formula indicate, a company's EBITA or NOPAT valuation multiple is driven by growth (g), ROIC, and the weighted average cost of capital (WACC). While most peers will have similar costs of capital, the other variables may be different, leading to differences in expected multiples.

A common flaw is to compare a particular company's multiple with an average multiple of other companies in the same industry, regardless of differences in their performance. Better to use a smaller subsample of peers with similar performance. Exhibit 18.11 shows the multiples of nine disguised companies that manufacture equipment and provide services for oil and gas drilling. The EV-to-NOPAT multiples range from approximately 10 times to almost 17 times. The company being evaluated, Swallow, had a multiple of 12 times, at the lower end of the range. Does this mean the company is undervalued? Probably not. When you examine the performance of the companies, you can see that they neatly divide into three groups: a top group with multiples of about 15 to 17 times, a middle group with multiples of about 12 times, and a low group with multiples of about 10 times. Note that the ROIC and growth

EXHIBIT 18.11 **Peer Groups by ROIC and Growth**



[1] Compound annual growth rate.

rates line up with the ranges of multiples. Swallow, with a multiple of 12 times, is valued right in line with the other two companies (Owl and Robin) that have similar ROIC and growth. If you didn't know Swallow's multiple, your best estimate would be the average of Owl and Robin, 12 times, not the average of the entire sample or some other sample.

Once you have collected a list of peers and measured their multiples properly, the digging begins. You must answer a series of questions: Why are the multiples different across the peer group? Do certain companies in the group have superior products, better access to customers, recurring revenues, or economies of scale? If these strategic advantages translate to superior ROIC and growth rates, better-positioned companies should trade at higher multiples.

ALTERNATIVE MULTIPLES

Although we have so far focused on enterprise value multiples based on EBITA or NOPAT, other multiples can prove helpful in certain situations. The EV-to-revenues multiple can be useful in bounding valuations with volatile EBITA. The P/E-to-growth (PEG) ratio somewhat controls for different growth rates across companies. Nonfinancial multiples can be useful for young companies where current financial information is not relevant. This section discusses each of these alternative multiples.

Enterprise Value to Revenues

In most cases, value-to-revenues multiples are not particularly useful for explaining company valuations, except in industries with unstable or negative profits. We'll use a simple example to illustrate. Companies A and B have the same expected growth, ROIC, and cost of capital; the only difference is that A's EBITA margin is 10 percent, while B's is 20 percent (B is more capital intensive, so its higher margin is offset by its greater invested capital). Because the companies have the same ROIC and growth, their value-to-EBIT ratios must be the same (13 times, based on the value driver formula). But the resulting value-to-revenues multiple is 1.3 for A and 2.6 for B. In this case, the value-to-revenues multiple tells us nothing about the valuations of the companies.

EV-to-revenues multiples are useful as a last resort in several situations. One is in the case of start-up industries, where profits are negative or a sustainable margin level can't be estimated. Another is in industries with highly volatile profit margins, where you believe that over the long term the companies will have roughly similar profit margins. You might also find situations where a company is periodically spending more on research and development (R&D) or marketing than its peers, so its earnings are temporarily depressed.

If investors are confident about the return to profit margins similar to those of peers, an EV-to-revenues multiple in line with peers might prove more relevant than an EV-to-EBITA multiple that is out of line with peers. Finally, a revenue multiple can provide a quick understanding of the potential value a company could generate if it were able to achieve the same levels of growth, operating margins, and capital efficiency as its peer group.

PEG Ratio

Some analysts and investors use a P/E-to-growth (PEG) ratio to assess the value of a company. For example, a company with a P/E of 15 and expected growth of 4 percent would have a PEG ratio of 3.75:

$$\text{PEG ratio} = \frac{\text{P/E}}{\text{Growth} \times 100} = \frac{15}{4\% \times 100} = 3.75$$

The PEG ratio is seriously deficient, however, because it doesn't take into consideration ROIC, which, as seen earlier, has a significant impact on a company's valuation.

While the concept of relating P/E to growth is relevant, there is no mathematical derivation that says you can simply divide one by the other and produce a significant result. Furthermore, there is no standardized approach for PEG ratios, particularly the choice of time horizon for growth. Should it be one year, five years, or a decade? The choice of horizon can make a big difference, as growth tends to flatten out over time. A company with 6 percent expected growth over five years may have only 4 percent expected growth over ten years. Shifting the growth horizon, in this case, would increase a company's PEG ratio by 50 percent. Finally, as you increase the time frame, growth rates in an industry will converge, so you will end up with differences in the PEG ratios just reflecting differences in P/Es.

The bigger problem, though, is ignoring ROIC. Exhibit 18.12 shows a DCF valuation we conducted for two companies. Company A has a higher ROIC (30 percent, versus 14 percent for B), while Company B has higher expected

EXHIBIT 18.12 **PEG Ratios Distorted by ROIC Differences**

	Company A	Company B
ROIC, %	30	14
Expected growth years 1–10, %	5	10
Expected growth after year 10, %	3	3
WACC, %	9	9
P/E = EV/NOPAT, times	17.0	17.0
PEG ratio, times	3.4	1.7

growth over the first ten years (10 percent, versus 5 percent for A). The DCF valuations of both companies at a 9 percent cost of capital and no debt lead to the same earnings multiple: 17 times. But Company A's PEG ratio is 3.4, while Company B's is 1.7. The common interpretation is that Company A is overvalued relative to Company B because its PEG ratio is higher. Yet it's clear that both companies are valued the same when both growth and ROIC are taken into account.

Multiples of Invested Capital

In some industries, multiples based on invested capital can provide better insights than earnings multiples. One example comes from the banking industry. In the years after the 2008 credit crisis, there was tremendous uncertainty about what levels of return on equity banks would be able to earn.[10] Furthermore, earnings forecasts one to three years out were not reliable and were often negative. Most investors resorted to using multiples of book equity. Banks with higher expected long-term returns on equity, based on their mix of businesses and the underlying economics of those businesses, tended to have higher multiples than banks in lower-return businesses. For example, banks whose portfolios emphasized wealth management and transaction processing, which are stable and earn high returns, were valued at higher multiples to equity than banks focused on more volatile and lower-return investment banking and retail banking.

Regulated industries provide another application of invested capital multiples. Under some regulatory regimes, profits are capped by the allowed return on a company's so-called regulatory asset base (RAB). The RAB is separately reported and represents the invested capital as calculated following certain rules that the regulator sets for qualified capital expenditures. If regulators were to not allow any excess returns above the cost of capital, the enterprise value-to-RAB multiple of a regulated company should be (close to) 1. In practice, the multiples end up at higher levels because regulators often provide various efficiency incentives allowing companies to generate excess returns. In addition, most companies have growth opportunities; they can expand their RAB by new, approved investment projects. For companies under similar regulatory regimes, many investors and analysts use RAB multiples for comparison and valuation.

Multiples Based on Operating Metrics

Sometimes company valuations are based on multiples of operating metrics. For example, values of oil and gas companies can be expressed as value per

[10] As explained in Chapter 38, we use return on equity, rather than return on capital, for banks.

barrel of oil reserves. Clearly, the amount of oil reserves in the ground the company has access to will drive the company's value. While the value of each barrel once the oil is extracted and sold is roughly the same, the costs to extract those barrels will vary widely and affect profit per barrel, depending on the geology of those reserves and the techniques needed to extract them. When estimating the value of an oil and gas company based on a valuation multiple of the amount of reserves it holds, you therefore have to make adjustments for any differences in the costs of extraction and distribution relative to the companies for which the multiple was estimated.

In other cases, investors and analysts resort to operating multiples when valuing young, fast-growing companies, because of the great uncertainty surrounding potential market size, profitability, and required investments. Financial multiples that normally provide a benchmark for valuation are often useless, as profitability (measured in any form) is often negative. A way to overcome this shortcoming is to apply nonfinancial multiples, comparing enterprise value with operating statistics such as website hits, unique visitors, or number of users or subscribers. This happened, for example, in the late 1990s, when numerous Internet companies went public with meager sales and negative profits. In 2000, *Fortune* reported market-value-to-customer multiples for a series of Internet companies.[11] *Fortune* determined that Yahoo was trading at $2,038 per customer, Amazon.com at $1,400 per customer, and NetZero at $1,140 per customer.

Today, similar multiples are sometimes used to analyze and compare the valuation of fast-growing companies with digital user–based or subscription-based business models. For example, the market value of audio- and video-streaming companies Netflix, Spotify, and Sirius XM can be expressed per paying user or subscriber. As of July 2019, Netflix was trading at about $1,200 per subscriber, Spotify at about $200 per subscriber, and Sirius XM at $900 per subscriber.[12] The question is whether such multiples offer real insights.

Effective use of a nonfinancial multiple requires that the nonfinancial metric be a reasonable predictor of future value creation and thus somehow tied to ROIC and growth. Simply taking the average of the customer or subscriber multiples for a set of apparently similar businesses provides little, if any, insight. Netflix, Spotify, and Sirius XM differ in terms of the underlying drivers of revenues and costs per user, because they operate with distinct business models. Netflix streams TV series, films, and documentaries on demand, and

[11] E. Schonfeld, "How Much Are Your Eyeballs Worth?" *Fortune*, February 21, 2000, 197–200.

[12] As of mid-2019, Netflix had an enterprise value of about $180 billion and about 150 million paying users, Spotify had a value of about $23 billion and 110 million paying users, and Sirius XM had a value of about $32 billion and 35 million paying users.

Spotify offers music streaming on demand. Sirius XM combines on-demand and radio-format music streaming (via its 2019 Pandora acquisition) with its original business of satellite radio broadcasting. All three generate revenues from user subscriptions, but Spotify and Sirius XM also generate advertising income. For Netflix, the cost of streamed content is largely independent of the number of users; it produces an increasing portion of the content itself and does not pay per view to its content suppliers. Spotify, in contrast, does not produce content and pays an amount per song-play to the content owners (record companies and artists). Sirius XM develops part of its content (for example, its radio shows) and buys music from third parties—also paying per song-play, but at lower rates for radio broadcasting than for on-demand streaming. In addition, the companies' user growth rates for 2019 varied widely: about 30 percent for Spotify, 20 percent for Netflix, and 3 percent for Sirius XM (with Sirius active in the United States only and Netflix and Spotify in many countries). Because of these differences in user economics and growth, an average of the value per user of Netflix, Spotify, and Sirius XM is not very meaningful for valuing another online music or video-streaming business. You would need to analyze the underlying business model in detail to understand which of the three is the most comparable business.

In the end, what matters is the underlying value creation, not the number of users, website hits, or unique visitors. In the late-1990s Internet examples from *Fortune*, Yahoo traded at a higher multiple than Amazon.com because investors expected that Yahoo's profit per user would be higher than Amazon's. Academic studies have demonstrated for these valuations that the number of unique visitors to a website or the number of pages on a site viewed per visit was directly correlated to a company's stock price, even after controlling for the company's current financial performance.[13] The power of a given nonfinancial metric, however, depended on the company. For portal and content companies such as Yahoo, page views and unique visitors were both correlated to a company's market value. For online retailers such as Amazon.com, only the page views per visit were correlated with value. Evidently, the market believed that merely stopping by would not translate to future cash flow for online retailers.

Research has also shown that as an industry matures, financial metrics such as gross profit and R&D spending become increasingly predictive, whereas nonfinancial data tend to lose power.[14] This indicates a return to traditional valuation metrics for new industries as they mature and once financial metrics became meaningful.

[13] B. Trueman, M. H. F. Wong, and X. J. Zhang, "The Eyeballs Have It: Searching for the Value in Internet Stocks," *Journal of Accounting Research* 38 (2000): 137–162.
[14] P. Jorion and E. Talmor, "Value Relevance of Financial and Non Financial Information in Emerging Industries: The Changing Role of Web Traffic Data" (working paper no. 021, London Business School Accounting Subject Area, 2001).

A problem with all multiples is that they are relative valuation tools. They measure one company's valuation relative to another's, normalized by some measure of size, be it size of earnings, revenues, or number of customers. They do not measure absolute valuation levels. For multiples based on operating metrics, there is an additional challenge of interpretation, because you can only compare them across a very limited number of companies that have a very similar operating model. Financial multiples are easier to interpret and compare. Take the example of an EV-to-EBITA multiple of 20 times for a mature industrial company. Basic understanding of underlying value drivers can readily lead you to a first conclusion: this multiple reflects high expectations for ROIC and growth (at a reasonable cost of capital). But it is much harder to come to such a conclusion when you observe an EV multiple of $1,200 per customer.

SUMMARY

Of the available valuation tools, discounted cash flow continues to deliver the best results. However, a thoughtful comparison of selected multiples for the company you are valuing with multiples from a carefully selected group of peers merits a place in your tool kit as well. When that comparative analysis is careful and well reasoned, it not only serves as a useful check of your DCF forecasts, but also provides critical insights into what drives value in a given industry. The distinction between operating and nonoperating results, capital, and cash flows should follow the exact same logic as applied in DCF valuation. The most insightful multiples are those that compare operating value to operating results. Operating metrics such as mineral reserve size or number of subscribers can be used when these are clearly related to value creation. In all cases, be sure that you analyze the underlying reasons that multiples differ from company to company, and never view multiples as a shortcut to valuation. Instead, approach your multiples analysis with as much care as you bring to your DCF analysis.

REVIEW QUESTIONS

1. DiversCo, a large U.S. company, operates in two areas: energy and retail clothing. You observe an analyst report that values DiversCo using multiples analysis based on a peer group of other large U.S. diversified companies. Do you think that this is an appropriate approach? Why or why not?
2. What is a forward-looking multiple? Why should one use forward-looking multiples as opposed to backward-looking multiples when valuing companies?
3. Exhibit 18.13 presents market and profit data for three companies that operate in the same industry. Using the data, compute enterprise value to EBITDA and enterprise value to EBITA for Companies 1 and 2. Is the net difference between Company 1 and Company 2 the same for both ratios? If not, why might this be?

Exhibit 18.13 **Multiples Analysis: Market and Profit Data**

$ million

	Company 1	Company 2	Company 3
Market data			
Share price$_t$, $	25	16	30
Shares outstanding$_t$, millions	5	8	15
Short-term debt$_t$	25	15	30
Long-term debt$_t$	50	70	40
Operating profit			
EBITDA$_{t+1}$	25	30	59
EBITA$_{t+1}$	22	23	51

4. Exhibit 18.13 presents market and profit data for three companies that operate in the same industry. If Company 3 has nonconsolidated subsidiaries valued at $50 million, what are the company's enterprise-value-to-EBITDA and enterprise-value-to-EBITA multiples? How might ignoring the impact of nonconsolidated subsidiaries affect your analysis of these three firms?
5. You are valuing multiple steady-state companies in the same industry. Company A is projected to earn $160 million in EBITA next year, grow at 2 percent per year, and generate ROICs equal to 15 percent. Company B is projected to earn $160 million in EBITA next year, grow at 6 percent per year, and generate ROICs equal to 10 percent. Both companies have an operating tax rate of 25 percent and a cost of capital of 10 percent. What are the enterprise-value-to-EBITA multiples for both companies? Does higher growth lead to a higher multiple in this case?

6. You are valuing multiple steady-state companies in the same industry. Company A is projected to earn $160 million in EBITA next year, grow at 2 percent per year, and generate ROICs equal to 15 percent. Company C is projected to earn $160 million in EBITA next year, grow at 5 percent per year, and generate ROICs equal to 12 percent. Both companies have an operating tax rate of 25 percent and a cost of capital of 10 percent. What are the enterprise-value-to-EBITA multiples for both companies? Does higher growth lead to a higher multiple in this case? Why do the results in Questions 5 and 6 differ?
7. Competitors BlueBev and RedBev have the same long-term prospects concerning growth and ROIC. RedBev temporarily stumbles during a new-product launch, and profits drop considerably as the company scrambles to fix the error. Which company is likely to have the higher enterprise-value-to-EBITA multiple? Why?
8. LeverCo is financed entirely by equity. The company generates an operating profit equal to $80 million. LeverCo currently trades at an equity value of $900 million. At a tax rate of 25 percent, what is the price-to-earnings multiple for LeverCo? New management decides to increase leverage through a share repurchase. The company issues $400 million in bonds to retire $400 million in equity. If the bonds pay interest at 5 percent, what is the company's new price-to-earnings ratio? Could you predict the direction the P/E ratio would move without performing this calculation?
9. What are the limitations of using enterprise-value-to-revenues multiples? When are they useful?